

Models	GSM8K	MATH
LLaMA2-7B: MetaMATH	66.6	19.2
+ RFT	68.5	19.9
+ ORM-PPO	70.8	20.8
+ MATH-SHEPHERD-step-by-step-PPO (Ours)	73.2	21.6
Mistral-7B: MetaMATH	77.9	28.6
+ RFT	79.0	29.9
+ ORM-PPO	81.8	31.3
+ MATH-SHEPHERD-step-by-step-PPO (Ours)	84.1	33.0

Table 2: Performances of different 7B models on GSM8K and MATH with greedy decoding. We use the questions in MetaMATH for RFT and PPO training. Both LLaMA2-7B and Mistral-7B are supervised by Mistral-7B-ORM and -MATH-SHEPHERD.

Baselines and Metrics In the verification scenario, following (Lightman et al., 2023), we evaluate the performance of our reward model by comparing it against the Self-consistency (majority voting) and outcome reward model. The accuracy of the best-of-N solution is utilized as the evaluation metric. For PRM, the minimum score across all steps is adopted to represent the final score of a solution. In the reinforcement scenario, we compare our step-by-step supervision with the outcome supervision provided by ORM, and Rejective Sampling Fine-tuning (RFT) (Yuan et al., 2023), we sample 8 responses for each question in MetaMATH for RFT. We use the accuracy of LLMs’ greedy decoding output to assess the performance.

4.1 MAIN RESULTS

MATH-SHEPHERD as verifier Table 1 presents the performance comparison of various methods on GSM8K and MATH. We find that: 1) As the verifier, MATH-SHEPHERD consistently outperforms self-consistency and ORM on two datasets with all generators. Specifically, enhanced by MATH-SHEPHERD, DeepSeek-67B achieves 93.3% and 48.1% accuracy on GSM8K and MATH; 2) In comparison to GSM8K, PRM achieves a greater advantage over ORM on the more challenging MATH dataset; This outcome aligns with the findings in Uesato et al. (2022) and Lightman et al. (2023). The former discovers that PRM and ORM yield similar results on GSM8K, whereas the latter shows that PRM significantly outperforms ORM on the MATH dataset. This could be attributed to the relative simplicity of the GSM8K dataset compared to MATH, i.e., the GSM8K dataset necessitates fewer steps for problem-solving. As a result, ORM operates efficiently when handling this particular dataset; 3) In GSM8K, when combined with self-consistency, there’s a drop in performance, whereas in MATH, performance improves. These results indicate that if the reward model is sufficiently powerful for a task, combining it with self-consistency may harm the verification performance.

MATH-SHEPHERD as reward model on reinforcement learning Table 2 presents the performance of different LLMs with greedy decoding outputs. As is shown: 1) step-by-step PPO significantly improves the performance of two supervised fine-tuned models. For example, Mistral-7B with step-by-step PPO achieves 84.1% and 33.0% on the GSM8K and MATH datasets, respectively; 2) RFT only slightly improves the model performance, we believe this is because MetaMATH already has conducted some data augmentation strategies like RFT; 3) the vanilla PPO with ORM can also enhance the model performance. However, it does not perform as well as the step-by-step PPO supervised by MATH-SHEPHERD, demonstrating the potential of step-by-step supervision.

MATH-SHEPHERD as both reward models and verifiers We also combine the reinforcement learning and the verification. As shown in Table 3: 1) reinforcement learning and verification are complementary. For example, in MATH, step-by-step PPO Mistral-7B outperforms supervised fine-tuning Mistral-7B 7.2% accuracy with self-consistency as the verifier; The performance gap is even larger than that of greedy decoding results, i.e., 4.4%; 2) after reinforcement learning, the vanilla verification methods with only reward models is inferior to self-consistency, we think the

Models	Verifiers	GSM8K	MATH500
Mistral-7B: MetaMATH	Self-Consistency	83.9	35.1
	ORM	86.2	36.4
	Self-Consistency + ORM	86.6	38.0
	MATH-SHEPHERD (Ours)	87.1	37.3
	Self-Consistency + MATH-SHEPHERD (Ours)	86.3	38.3
Mistral-7B: MetaMATH +step-by-step PPO (Ours)	Self-Consistency	87.4	42.3
	ORM	87.6	41.3
	Self-Consistency + ORM	89.0	43.1
	MATH-SHEPHERD (Ours)	88.4	41.1
	Self-Consistency + MATH-SHEPHERD (Ours)	89.1	43.5

Table 3: Results of reinforcement learning and verification combination. The reward models are trained based on Mistral-7B. The verification is based on 256 outputs.

reason is that the initial reward model is not sufficient to supervise the more powerful model after PPO. These results can also show the potential of iterative reinforcement learning, which we leave for future work.

5 ANALYSIS

5.1 PERFORMANCE WITH DIFFERENT NUMBER OF CANDIDATE SOLUTIONS

Figure 3 illustrates the performance comparison of various strategies when applied to different numbers of candidates ranging from 1 to 256 on two benchmarks. The key observations are as follows: 1) PRM exhibits consistent superior performance when compared to both ORM and majority voting, with the degree of this superiority becoming more pronounced as N escalates. 2) In MATH, our automatically annotated datasets outperform the human-annotated PRM800K (Lightman et al., 2023). We ascribe this superiority to the distribution gap and the data quantity. Specifically, PRM800K is annotated based on the output from GPT-4, and consequently, a discrepancy arises for the output of open-source LLaMA models fine-tuned on MetaMATH. Furthermore, when considering the quantity of data, our automated reward model data exhibits both high scalability and a reduced labeling cost. Consequently, our dataset is four times larger than that provided in PRM800K. Overall, these results further underscore the effectiveness and potential of our method.

5.2 QUALITY OF THE AUTOMATIC PROCESS ANNOTATIONS

In this section, we explore the quality of our automatic PRM dataset. To achieve this, we manually annotate 160 steps sampled from the training set of GSM8K and use different completers to infer from each step to achieve their label. We find that:

Automatic process annotation exhibits satisfactory quality. Figure 4(a) demonstrates that utilizing LLaMA2-70B trained on MetaMATH as the completer, the accuracy of the hard estimation (HE) reaches 86% when N equals 4. This suggests that our automatically constructed dataset is of high quality. However, we observed a decline in the accuracy of the constructed dataset with further increases in N . Our analysis indicates that larger values for N may lead to false positives.

Figure 4(b) shows the cross-entropy loss between SE and HE labels compared to the human-annotated distribution: as N increases, SE progressively aligns closer to the standard distribution, in contrast to HE which does not exhibit similar behavior. It is essential to note that at $N=4$, HE achieves an accuracy of 86%. We can theoretically attain higher quality data exceeding 86% accuracy by utilizing SE. However, we discovered that the performance of the verifier exhibits no substantial divergence whether trained with either SE or HE. This may be attributable to the already high-quality annotations provided by HE.

Furthermore, we also delve into other automatic process annotation methodologies. For instance, (Li et al., 2023b) employs a natural language inference (NLI) model and a string match rule to annotate a

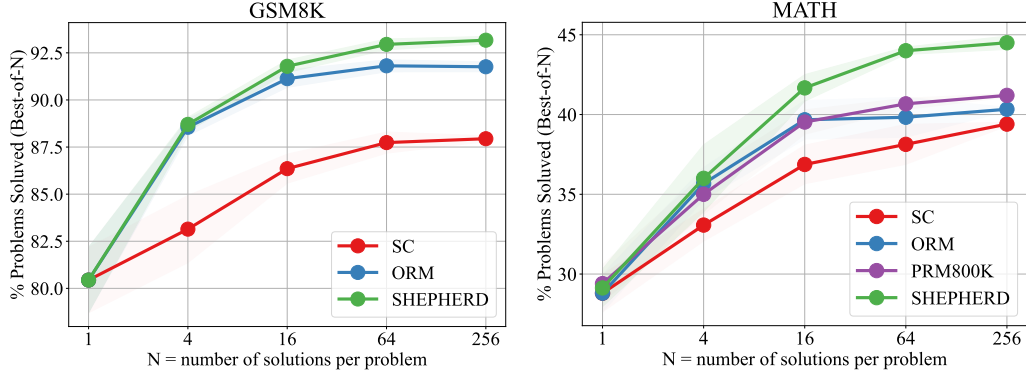


Figure 3: Performance of LLaMA2-70B using different verification strategies across different numbers of solution candidates on GSM8K and MATH.

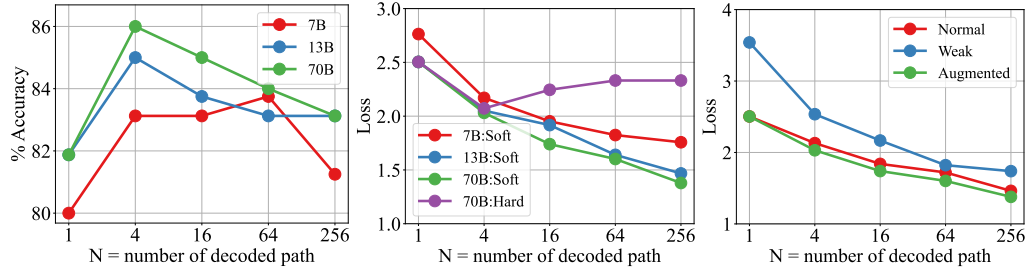


Figure 4: Quality of process annotation on GSM8K. (a): Accuracy of the process annotation using different completer; (b): Loss of the process annotation using different completer; (c): Loss of the process annotation using the same completer with different training data.

given step. The NLI-based method annotates a step as correct if it is entailment with any step in the reference solutions. The Rule-based method annotates a step as correct if its support number precisely matches that of any steps in the reference solutions. As demonstrated in Table 4, our annotation strategy exhibits substantial superiority over the two approaches.

The ability of the LLM completer plays an important role in the data quality. We employ a completer to finalize multiple subsequent reasoning processes for a given step. Therefore, we investigate the impact of the LLM completer.

Figure 4(b) presents the cross-entropy loss across diverse completers trained on MetaMath. The results indicate that a larger completer is adept at generating superior-quality datasets. Figure 4(c) depicts the cross-entropy loss of LLaMA2-70B trained with different datasets. ‘Normal’ denotes the original GSM8K training dataset; ‘Weak’ refers to the Normal set excluding examples whose questions are in our 160 evaluation set; while ‘Augmented’ symbolizes MetaMath, an augmented version of the Normal set.

The findings suggest that high-quality training sets allow the model to operate more proficiently as a completer. Importantly, the ‘Weak’ set exhibits a markedly larger loss than other datasets. This insight drives us to infer that LLMs should acquire the questions in advance to enhance their performance as completers. We can also conjecture that a stronger foundational model, coupled with superior training data, could further enhance the quality of automatic annotation.

5.3 INFLUENCE OF THE PRE-TRAINED BASE MODELS

To conduct an exhaustive evaluation of MATH-SHEPHERD’s effectiveness, we performed a diverse range of experiments using model sizes 7B, 13B, and 70B.